

MUHAMMAD ABDUL FATTAH

Software Engineer | Mobile Payments, Backend APIs, Reliability

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PROFESSIONAL PROFILE

Software Engineer with 4+ years of overall experience across reliability-sensitive payment systems, mobile engineering, backend API integration, and production delivery. At BCA, support Android POS merchant payment flows serving 15,000+ transactions per minute and have investigated or resolved 50+ critical production issues. Strongest in Kotlin/Java, transaction-sensitive client behavior, SQL-backed debugging, CI/CD, and operational reliability, with public hands-on projects in Spring Boot payment correctness, observability, and secure Android engineering.

CORE SKILLS

Production-backed software: Kotlin, Java, Android SDK, MVVM, RxJava, AIDL/Binder exposure, REST APIs, backend API integration, Android service behavior, and secure mobile controls.

Production-backed delivery and reliability: Payment and POS systems, transaction-state investigation, settlement-related controls, SQL-backed debugging, Git, Jenkins, CI/CD, Dynatrace, Elasticsearch, incident investigation, code review, and release discipline.

Public-project-backed and actively deepening: Spring Boot, automated test and CI patterns, Kafka and WebSocket exposure, Docker, GitHub Actions, Prometheus, Grafana, Helm, Kubernetes exposure, PostgreSQL, OpenAPI/Swagger, Go, FastAPI, Python, and cloud-native API patterns.

PROFESSIONAL EXPERIENCE

PT Bank Central Asia (BCA) | *IT Specialist - Software Engineering / Payments* | Tangerang, Indonesia | Nov 2023 - Present

- Work on APOS BCA, a reliability-sensitive Android POS merchant payment platform supporting 15,000+ transactions per minute, with Kotlin/Java transaction flows, backend API integration, Android service behavior, and controlled release practices for merchant operations.
- Support settlement-related transaction controls and reporting for merchant payment flows, helping distinguish eligible completed transactions from unresolved cases requiring investigation or reversal handling.
- Investigated or resolved 50+ critical production issues by tracing device and service behavior, API responses, transaction state, release context, Dynatrace telemetry, and Elasticsearch logs to move recurring issues beyond symptom-level fixes.
- Maintain and debug merchant-client behavior across Kotlin/Java Android components, MVVM, RxJava, and Android service lifecycle for reliability-sensitive merchant operations.
- Investigate terminal-level integration behavior across AIDL/Binder exposure, USB/ECR-linked terminal environments, and third-party or vendor SDK behavior.
- Contribute to Java/Spring Boot backend integration, SQL-backed investigation, Kafka and WebSocket exposure, and Jenkins plus PCF/Cloud Foundry-related delivery or runtime workflows.
- Work within Git-based branch, review, merge, and deployment discipline; document technical findings and coordinate directly with product, QA, security, infrastructure, operations, and business stakeholders on risk-sensitive changes.
- Support secure mobile-client behavior through root and emulator detection work, DexGuard/ProGuard hardening discussions, and transaction-sensitive failure analysis.

ADDITIONAL SOFTWARE EXPERIENCE

Telkom Indonesia | *iOS Engineer Intern* | Indonesia | Apr 2023 - Sep 2023

- Contributed to SwiftUI SDK work connected to approximately 10 government applications, with attention to maintainable mobile integration, SDK delivery, and cross-team coordination.

Apple Developer Academy Indonesia | *Junior iOS Developer / Apprentice* | Indonesia | Feb 2022 - Dec 2022

- Shipped 4 application prototypes end-to-end in Scrum teams, from product discovery and implementation through testing and feedback-driven iteration; led 2 teams when needed.
- Worked with designers, developers, and mentors to translate ambiguous needs into working product increments and clear demos.

PT Orbit Ventura Indonesia | *AI Developer* | Indonesia | Sep 2021 - Feb 2022

- Built backend services and REST APIs for AI inference workflows, connecting model integration, API behavior, and delivery requirements across the product flow.
- Contributed to a project that won 1st Place at the National AI Project Award 2021.

SELECTED PROJECTS

PayFlow Reliability | *Spring Boot Backend Reliability Lab* | github.com/fattah247/payflow-reliability

- Built a public-safe Java/Spring Boot fintech sandbox around idempotent payment creation, duplicate webhook handling, guarded payment-state transitions, settlement batching, reconciliation mismatch detection, manual review flows, and audit events.
- Included automated tests, CI, demo evidence, and documentation to make transaction-correctness and failure-handling decisions inspectable without using employer code, data, architecture, or workflows.

iYup | *Self-Hosted Uptime & Latency Monitoring Lab* | github.com/fattah247/iYup

- Built a Go ping/check service with FastAPI status APIs, Prometheus metrics, Grafana dashboards, Alertmanager wiring, Docker Compose, Helm rendering, verification scripts, and CI to demonstrate observability and operational visibility.

TrustGate Android | *Kotlin Mobile Security Lab* | github.com/fattah247/trustgate-android

- Built device-risk checks, root/emulator/debugger signal handling, sensitive-action gating, HMAC request signing, encrypted local storage, certificate-pinning tradeoffs, security event logs, unit tests, Android CI, and documentation.

Stock-Triage | *Local-First Financial Analysis Workflow* | github.com/fattah247/Stock-Triage

- Built a local-first workflow for data extraction, parsing, and source-backed AI-assisted analysis, emphasizing evidence and reviewable output over ungrounded model claims.

PhotoReviewApp | *iOS Photo Review Application*

- Built an iOS product prototype using SwiftUI, Core Data, and Photos integration, with deep links, widgets, and analytics to support a practical photo-review workflow.

EDUCATION & CERTIFICATIONS

University of Indonesia | *Bachelor of Engineering, Biomedical Engineering* | GPA 3.67 / 4.00

- Thesis work covered alginate/hydroxyapatite/collagen scaffold fabrication with turmeric and ginger extracts, including freeze-drying, UV-Vis, DPPH, MTT exposure, UC-MSC viability/proliferation testing, hemocytometer work, microscopy, and two-way ANOVA.

AWS Certified Cloud Practitioner | AWS Certified Solutions Architect - Associate: In Progress

ADDITIONAL INFORMATION

English: professional working proficiency for technical documentation, written communication, and meetings. Based in Indonesia; open to remote, contract, and relocation opportunities.